25900870 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCIAM100 - INTERACTIVE ROBOTIC SYSTEMS

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which pin on an Arduino Uno board is used for SPI communication? [K] / [CO1]

   a). pin10 b). pin11 c). pin12 d). pin13

2. What is the purpose of the" tone "function? [U] / [CO1]

   a). generate a tone on a pin b). generate a pulse on a pin c). read an analog value d). write a digital value

3. What language is the Arduino IDE built on? [U] / [CO1]

   a). Java b). HTML c). C/C++ d). Python

4. IR flame sensor senses _____ emitted by flames [U] / [CO2]

   a). Infrared Radiation b). UV Radiation c). X Ray d). None of these

5. The principle of operation of MQ 2 smoke sensor is the change of _____ in the presence of gas [U] / [CO2]

a). voltage b). resistance c). current d). capacitance

6. Number of pins present in HC-SR04 [K] / [CO2]

a). 3 b). 4 c). 5 d). 6

7. Which technique involves predicting future states based on current data to optimize control actions in self-balancing robots? [U] / [CO3]

a). Feedforward Control b). Adaptive Control c). PID Control d). Hysteresis Control

8. How many number of pins does the MPU6050 sensor typically have? [K] / [CO3]

a). 4 b). 6 c). 8 d). 10

9. What is the typical output range of a standard servo motor when controlled via PWM? [K] / [CO3]

a). 0-90 degrees b). 0-180 degrees c). 0-360 degrees d). 0-45 degrees

10. What are the potential applications of a radar mode system using an ultrasonic sensor and servo motor? [U] / [CO3]

a). Obstacle detection b). Object tracking c). Proximity sensing d). All of the above

[10x1 = 10]

Part B

Short answer Type Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Explain how you would Troubleshoot an Arduino program that is not executing the void loop () function as expected. What steps would you take to identify and fix the issue. [K] / [CO1]

12. What are the different mathematical operators used in Ardino programming [U] / [CO1]

13. What is the pourpose of an H bridge circuit ? [U] / [CO2]

14. Why LDR is connected to analog pin instead of digital pin of arduino ? [U] / [CO2]

15. What are the implications of using the HC-SR04 sensor in outdoor environments, particularly in terms of weather conditions? [U] / [CO3]

16. Define infrared (IR) sensors and explain their basic operating principle in obstacle detection. [K] / [CO3]

[4x5=20]

Part C

Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

17.	Write a program that uses a WHILE loopto blink an LED 5 times, with a 500ms delay between each blink?	[K] / [CO1]
18.	Explain the internal structure and parts of a DC in detail?	[U] / [CO2]
19.	Write a program to rotate a servo from 0 Degre to 180 Degree, 5 Times	[U] / [CO2]
20.	Investigate the influence of terrain and environmental conditions on the direction control of robotic vehicles. How do factors such as slope, surface texture, and obstacles affect navigation strategies?	[QNSMARTI:T:B20] / [CO3]

[2x10 =20]